

Probability Inequalities

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This note summarizes some useful probability inequalities: the Markov inequality (Section 1), Chebyshev's inequality (Section 2), Cantelli's inequality (Section 3), Hoeffding's inequality (Section 4), Chernoff bound (Section 5), and Jensen's inequality (Section 6).

1 Markov Inequality

Let X be a nonnegative random variable and $a > 0$. Then we have

$$\Pr[X \geq a] \leq \frac{\mathbb{E}[X]}{a}.$$

Proof. Intuitively,

$$\mathbb{E}[X] = \Pr[X < a] \cdot \mathbb{E}[X \mid X < a] + \Pr[X \geq a] \cdot \mathbb{E}[X \mid X \geq a].$$

Note $\mathbb{E}[X \mid X \geq a] \geq a$ and $\mathbb{E}[X \mid X < a] > 0$ since $X \geq 0$. Therefore,

$$\mathbb{E}[X] \geq a \Pr[X \geq a].$$

Another approach: Let $f(x)$ the pdf of X ,

$$\mathbb{E}[X] = \int_0^\infty x f(x) dx \geq \int_a^\infty x f(x) dx \geq \int_a^\infty a f(x) dx = a \Pr[X \geq a].$$

□

Corollary 1. Given a monotonic ϕ ,

$$\Pr[X \geq a] = \Pr[\phi(X) \geq \phi(a)] \leq \frac{\mathbb{E}[\phi(X)]}{\phi(a)}.$$

2 Chebyshev's Inequality

Let X be a random variable and $a > 0$. Then we have

$$\Pr[|X - \mathbb{E}[X]| \geq a] \leq \frac{\text{Var}[X]}{a^2}.$$

One also writes for $k \geq 1$,

$$\Pr[|X - \mathbb{E}[X]| \geq k\sigma] \leq \frac{1}{k^2}.$$

Proof. Let $Y = (X - \mathbb{E}[X])^2$. With constant a^2 , the Markov inequality gives

$$\Pr[Y \geq a^2] \leq \frac{\mathbb{E}[Y]}{a^2}.$$

Note that $Y \geq a^2 \Leftrightarrow |X - \mathbb{E}[X]| \geq a$, and $\mathbb{E}[Y] = \text{Var}[X]$.

□

Corollary 2.

1. *Multidimensional Chebyshev's inequality*

$$\Pr \left[\sum_{i=1}^n (X_i - \mu_i)^2 \geq k^2 \sum_{i=1}^n \sigma_i^2 \right] \leq \frac{1}{k^2}.$$

Or

$$\Pr[\|X - \mu\| \geq k\|\sigma\|] \leq \frac{1}{k^2}.$$

2. *Higher Moments: By applying Markov's inequality to nonnegative $|X - \mathbb{E}[X]|^n$,*

$$\Pr \left[|X - \mathbb{E}[X]| \geq k \mathbb{E}[|X - \mathbb{E}[X]|^n]^{\frac{1}{n}} \right] \leq \frac{1}{k^n},$$

where when $n = 2$, it is the Chebyshev's inequality.

3. *One-tailed Chebyshev's inequality*

$$\Pr[X - \mu \geq k\sigma^2] \leq \frac{1}{1 + k^2}.$$

This comes from the Cantelli's inequality.

3 Cantelli's Inequality

Let X be a random variable and $\lambda > 0$. Then we have

$$\Pr[X - \mathbb{E}[X] \geq a] \leq \frac{\sigma^2}{\sigma^2 + a^2}.$$

One also has

$$\Pr[X - \mathbb{E}[X] \leq -a] \leq \frac{\sigma^2}{\sigma^2 + a^2}.$$

Proof. Let $Y = X - \mathbb{E}[X]$, and thus $\mathbb{E}[Y] = 0$ and $\text{Var}(Y) = \sigma^2$.

$$\Pr[Y \geq a] = \Pr[Y + u \geq a + u] \leq \Pr[(Y + u)^2 \geq (a + u)^2] \leq \frac{\mathbb{E}[(Y + u)^2]}{(a + u)^2} = \frac{\sigma^2 + u^2}{(a + u)^2}.$$

for any $u \in \mathbb{R}$. We can choose u to make the last term minimal by differentiation. That is, $u = \sigma^2/a$. Now,

$$\Pr[Y \geq a] \leq \frac{\sigma^2 + u^2}{(a + u)^2} = \frac{\sigma^2}{a^2 + \sigma^2}.$$

□

4 Hoeffding's Inequality

Let $X_1, X_2, \dots, X_n$ be independent random variables and $a_i \leq X_i \leq b_i$ almost surely. Let $S_n = \sum_{i=1}^n X_i$. The Hoeffding's inequality states that for all $t > 0$,

$$\Pr[S_n - \mathbb{E}[S_n] \geq t] \leq \exp \left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2} \right).$$

Equivalently,

$$\Pr[|S_n - \mathbb{E}[S_n]| \geq t] \leq 2 \exp \left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2} \right).$$

Special Case: If $a_i = 0, b_i = 1$, the Hoeffding's inequality gives

$$\Pr[S_n - \mathbb{E}[S_n] \geq t] \leq \exp \left(-\frac{2t^2}{n} \right).$$

Lemma 1 (Hoeffding's Lemma). *Let X be any random variable and $a \leq X \leq b$ almost surely. For any $s \in \mathbb{R}$,*

$$\mathbb{E}[e^{s(X - \mathbb{E}[X])}] \leq \exp \left(\frac{s^2(b - a)^2}{8} \right).$$

Proof. We may assume WLOG that $\mathbb{E}[X] = 0$, otherwise we can replace X with $X - \mathbb{E}[X]$. Since e^{sx} is convex, we have

$$e^{sx} \leq \frac{b-x}{b-a} e^{sa} + \frac{x-a}{b-a} e^{sb}.$$

$$\mathbb{E}[e^{sX}] \leq \frac{b - \mathbb{E}[X]}{b-a} e^{sa} + \frac{\mathbb{E}[X] - a}{b-a} e^{sb} = \frac{b}{b-a} e^{sa} + \frac{-a}{b-a} e^{sb} = \exp[L(\lambda(b-a))],$$

where

$$L(h) = \frac{ha}{b-a} + \ln \left(1 + \frac{a - e^h a}{b-a} \right).$$

By Taylor's theorem, there exists some $0 \leq \theta \leq 1$,

$$L(h) = L(0) + hL'(0) + \frac{1}{2}h^2L''(h\theta) = -\frac{1}{2}h^2 \frac{abe^{h\theta}}{(b - ae^{h\theta})^2} \leq \frac{1}{8}h^2,$$

where we use AMGM inequality to derive $L''(h) \leq \frac{1}{4}$. In conclusion,

$$\mathbb{E}[e^{sX}] \leq \exp \left(\frac{1}{8}s^2(b-a)^2 \right).$$

□

Proof of Hoeffding's Inequality. Using Hoeffding's lemma, we can prove the Hoeffding's inequality. First from Markov's inequality and independence of X_i , for any $s, t > 0$,

$$\begin{aligned} \Pr[S_n - \mathbb{E}[S_n] \geq t] &= \Pr[\exp(s(S_n - \mathbb{E}[S_n])) \geq \exp(st)] \\ &\leq \frac{\mathbb{E}[\exp(s(S_n - \mathbb{E}[S_n]))]}{\exp(st)} \\ &\leq \frac{\prod_{i=1}^n \mathbb{E}[\exp(s(X_i - \mathbb{E}[X_i]))]}{\exp(st)}. \end{aligned}$$

By Hoeffding's lemma,

$$\begin{aligned} \mathbb{E}[\exp(s(X_i - \mathbb{E}[X_i]))] &\leq \exp \left(\frac{1}{8}s^2(b_i - a_i)^2 \right), \\ \implies \Pr[S_n - \mathbb{E}[S_n] \geq t] &\leq \exp \left(-st + \frac{1}{8}s^2 \sum_{i=1}^n (b_i - a_i)^2 \right). \end{aligned}$$

The upperbound is minimized when

$$s = \frac{4t}{\sum_{i=1}^n (b_i - a_i)^2}.$$

Hence,

$$\Pr[S_n - \mathbb{E}[S_n] \geq t] \leq \exp \left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2} \right).$$

□

5 Chernoff Bound

Let X be a random variable and $t > 0$, by Markov's inequality

$$\Pr[X \geq a] = \Pr[e^{tX} \geq e^{ta}] \leq \frac{\mathbb{E}[e^{tX}]}{e^{ta}}.$$

Note this applies for all t ,

$$\Pr[X \geq a] \leq \inf_{t \geq 0} \frac{\mathbb{E}[e^{tX}]}{e^{ta}}.$$

5.1 Application

Let X_i be Bernoulli random variable with probability p_i , and let $X = \sum X_i$. First note that

$$\mathbb{E}[e^{tX_i}] = 1 - p_i + p_i e^t = 1 + p_i(e^t - 1).$$

Consider the inequality $1 + y \leq e^y$, $\mathbb{E}[e^{tX_i}] \leq e^{p_i(e^t - 1)}$, and thus

$$\mathbb{E}[e^{tX}] \leq \prod e^{p_i(e^t - 1)} = e^{\mu(e^t - 1)},$$

where $\mu = \mathbb{E}[X]$. Now let $a = (1 + \delta)\mu$, $t = \ln(1 + \delta)$, we have

$$\Pr[X \geq (1 + \delta)\mu] \leq (1 + \delta)^{-(1 + \delta)\mu} e^{\mu\delta} = \left(\frac{e^\delta}{(1 + \delta)^{(1 + \delta)}} \right)^\mu.$$

Similarly,

$$\Pr[X < (1 - \delta)\mu] \leq (1 + \delta)^{-(1 - \delta)\mu} e^{-\mu\delta} = \left(\frac{e^{-\delta}}{(1 - \delta)^{(1 - \delta)}} \right)^\mu.$$

One step further, logarithm of the RHS gives

$$\mu(\delta - (1 + \delta) \ln(1 + \delta)).$$

By the inequality $\ln(1 + x) \leq \frac{x}{1 + x/2}$, we have

$$\mu(\delta - (1 + \delta) \ln(1 + \delta)) \leq \mu(\delta - (1 + \delta) \frac{\delta}{1 + \delta/2}) = -\frac{\delta^2 \mu}{2 + \delta}.$$

Therefore,

$$\Pr[X \geq (1 + \delta)\mu] \leq \exp\left(-\frac{\delta^2 \mu}{2 + \delta}\right) \quad \forall \delta > 0 \quad (1)$$

$$\Pr[X \leq (1 - \delta)\mu] \leq \exp\left(-\frac{\delta^2 \mu}{2}\right) \quad \forall \delta \in (0, 1) \quad (2)$$

Moreover,

$$\Pr[|X - \mu| \geq \delta\mu] \leq 2e^{-\mu\delta^2/3} \quad \forall \delta \in (0, 1) \quad (3)$$

6 Jensen's Inequality

First define a convex function.

Definition 1 (Convex Function). A function $f(x)$ is said to be convex over interval (a, b) if for all $x_1, x_2 \in (a, b)$ and $0 \leq \lambda \leq 1$,

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2).$$

It is called strictly convex if the equality holds only if $\lambda = 0$ or $\lambda = 1$. A function is concave if $(-f)$ is convex.

Theorem 1 (Non-negative Second Derivative). *If a function $f(x)$ has a second derivative that is non-negative (positive) over an interval, then $f(x)$ is (strictly) convex over that interval.*

Proof. By Taylor's expansion, let x_0 be some point in the interval,

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x^*)}{2}(x - x_0)^2,$$

where x^* lies between x and x_0 . Since second derivative is non-negative, we have

$$f(x) \geq f(x_0) + f'(x_0)(x - x_0).$$

Let $x_0 = \lambda x_1 + (1 - \lambda)x_2$. Take $x = x_1$, we obtain

$$f(x_1) \geq f(x_0) + f'(x_0)((1 - \lambda)(x_1 - x_2)). \quad (4)$$

Take $x = x_2$, we obtain

$$f(x_2) \geq f(x_0) + f'(x_0)(\lambda(x_2 - x_1)). \quad (5)$$

Multiply Equation (4) by λ and (5) by $1 - \lambda$, and adding them, we obtain

$$\lambda f(x_1) + (1 - \lambda)f(x_2) \geq f(x_0) = f(\lambda x_1 + (1 - \lambda)x_2).$$

□

Theorem 2 (Jensen's Inequality). *If f is a convex function and X is a random variable, then*

$$\mathbb{E}[f(X)] \geq f(\mathbb{E}[X]).$$

If f is strictly convex, then the equality implies $X = \mathbb{E}[X]$ with probability 1 (X is a constant).

Proof. We only prove for discrete case. By induction on the size of support. For two-point support, with probability p_1, p_2 , by convexity,

$$\mathbb{E}[f(X)] = p_1 f(x_1) + p_2 f(x_2) \geq f(p_1 x_1 + p_2 x_2) = f(\mathbb{E}[X]).$$

Now suppose it is true for support of $k - 1$ points. Let $p'_i = p_i / (1 - p_k)$ for $i = 1, 2, \dots, k - 1$, we have

$$\begin{aligned} \mathbb{E}[f(X)] &= p_k f(x_k) + (1 - p_k) \sum_{i=1}^{k-1} p'_i f(x_i) \\ &\geq p_k f(x_k) + (1 - p_k) f\left(\sum_{i=1}^{k-1} p'_i x_i\right) \\ &\geq f\left(p_k x_k + \sum_{i=1}^{k-1} p'_i x_i\right) = f(\mathbb{E}[X]). \end{aligned}$$

□